Minimum-time S-curve Commands for Vibration-free Transportation of an Overhead Crane with Actuator Limits

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Abstract

S-curve commands are ubiquitous in servo drives owing to their simplicity and smoothness. Nevertheless, they need to be adapted for use in flexible systems, where the problem of residual vibration must be addressed. This paper proposes a simple motion planning method for the vibration-free transfer process of an overhead crane using S-curve commands. Based on a position baseline S-curve, which is generated from a bang-off-bang acceleration profile, two approaches are proposed to build the vibration suppression capability. One is an embedding method that injects the essential terminal conditions for vibration-free transportation into the baseline S-curve command without altering its original form. The other is a shaping method inspired from the input shaping technique. In both schemes, the baseline S-curve is parameterized to establish minimum-time optimization problems, in which maximum velocity and maximum acceleration of the actuator are explicitly taken into consideration. The minimum-time solutions are successfully obtained by solving constrained (discrete) nonlinear programs. An online trajectory generation can be realized using the proposed approach. Both simulation and experimental results are given to verify the effectiveness of the proposed method.

Keywords: Vibration control, Crane control, S-curve, Input shaping, Motion planning, Underactuated system.

1. Introduction

Overhead cranes are widely considered the most important means of transportation in various places such as factories, harbors, etc. to transfer heavy payloads point-to-point within the workspace. The payload is suspended under the trolley by rope and its height can be adjusted by a hoisting motion. The trolley actuator relocates the payload. This is always a linear motion, and for this reason, the overhead crane is usually described in a Cartesian coordinate. As the rope is flexible, the payload exhibits residual vibration during and after transportation. This undesirable oscillation significantly reduces the safety and productivity of the overhead crane system in practice. Consequently, the control objectives for this system are bringing the trolley to its destination and suppressing the swing movement of the payload. This is a challenging task since the overhead crane is an underactuated system—it has only one control input (trolley actuation) and two control outputs. Therefore, considerable interests were mounting on the vibration control for the overhead crane system. They can be classified into two main categories: feedback and feedforward controls. These types can be combined to obtain a 2-DOF control system to improve control performance.

An extensive background can be found in the feedback control category. Naturally, the vibration control of overhead cranes has evolved along with the development of control theory. New designs of the linear controls and model predictive controls have been promptly employed on the overhead cranes (Sawodny et al., 2002; Mori and Tagawa, 2018; Schlott et al., 2019; Š. Ileš et al., 2018; Galuppini et al., 2018). Feedback linearization and differential-flatness based methodology (Rigatos, 2015), which aims at transforming a nonlinear model to a linear one to utilize well-established linear controls, have also been widely applied (Tuan et al., 2013; Wu and He, 2016). The main disadvantage of the feedback linearization method is that it is not robust to parametric uncertainties and unmodeled dynamics. In order to deal with such circumstances, sliding-mode based controls (Almutairi and Zribi, 2009; Hieu and Hong, 2012; Tho and Terashima, 2019), adaptive controls (Fang et al., 2012; Sun et al., 2015; Zhang et al., 2017; Yang et al., 2018), energy and passivity based nonlinear controls (Sun et al., 2019; Lu et al., 2018; Chen and Sun, 2019) are preferable. Intelligence—fuzzy and neural network—controls have also been successfully implemented on crane systems (Lee et al., 2014; Yang et al., 2019; Qiu and Zhang, 2019). To sum up, feedback control techniques show excellent properties in handling parametric uncertainties and external disturbances. However, they demand a sophisticated and costly sensor structure that requires regular maintenance, and thus feedback controllers are still not popular

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in practical cranes. In numerous contexts, feedforward controls are preferable in industry due to their simplicity and ease of implementation, as well as no additional measurement system being required.

Minimum-time trajectory planning is a crucial means to enhance transportation productivity of overhead crane systems. To obtain a universal minimum-time Zero Vibration (ZV) reference trajectory in the presence of both state and control input constraints, one can employ indirect (Auernig and Troger, 1987; Terashima et al., 2007; Tho et al., 2018, 2019) or direct methods (Cruz and Leonardi, 2013; Zhang et al., 2014; Alghanim et al., 2015; Kolar et al., 2017). Indirect approaches utilize the necessary condition for time-optimality based on the Pontryagin Minimum Principle, whereas direct methods discretize the system to formulate and solve linear/nonlinear programs by means of optimization solvers. However, the above-mentioned techniques require high computational effort, and thus they can barely generate motion trajectories in an online fashion. To address this issue, one can limit the attention to a certain class of control inputs and establish an analytical minimum-time solution for that class. In this way, an online trajectory generation can be achieved. The class of bang-off-bang acceleration control inputs (Meckl and Arestides, 1998; Sun et al., 2012; Hoang et al., 2014) is the most basic and commonly used in industrial servo drives compared to other classes such as B-splines (Chen et al., 2016), trigonometric (Sun et al., 2012; Uchiyama et al., 2013), jerk-limited (Béarée, 2014), etc. In Meckl and Arestides (1998), a S-curve position command generated from a bang-off-bang acceleration profile was proposed to suppress vibration for a flexible system. The excitation energy of the input forcing function was minimized by using frequency analysis. Unfortunately, the actuator limits were omitted in the design process. Recently, Sun et al. (2012) proposed a three-segment (i.e. bang-off-bang) acceleration profile and its modified versions for a 2D overhead crane. A geometric approach, based on the system phase portrait, is employed to derive vibration suppression conditions for the S-curve profile. Hoang et al. (2014) introduced a similar approach as Sun et al. (2012) and extended those results by considering a family of acceleration profiles in a stair form. A minimum-time solution, however, was not considered in the above-mentioned studies. In summary, to date, none of the existing studies have solved the minimum transferring time problem for ZV S-curve commands generated from the class of bang-off-bang acceleration profiles under restrictions imposed by actuator limitations.

The present study aims to obtain a low-cost crane system with enhanced productivity, while maintaining a precise payload positioning and the vibration suppression capability. Therefore, in response to the previous discussions, this paper will establish minimum-time solutions for the ZV S-curve position commands for an overhead crane system considering both maximum velocity and maximum acceleration of the actuator. In this paper, the S-curve command is understood in the position level and it is created from a bang-off-bang acceleration trajectory. Two approaches will be introduced to enable the S-curve to admit the vibration cancellation capability. The embedding method directly injects the vibration suppression conditions at the end of maneuver into the baseline S-curve command. Consequently, the resultant S-curve command—named S-curve–Vibration Suppression Embedded (SC–VSE)—has internal constraints on the duration of acceleration, uniform, and deceleration intervals that strongly relate to the natural frequency of the system. On the other hand, the shaping method does not impose those internal constraints on the baseline S-curve but will use some add-on tools to modulate the original (baseline/unshaped) S-curve command to gain vibration-free transportation. In this paper, the most well-known add-on—ZV input shaper (Singer et al., 1997; Singhose et al., 2000; Pereira et al., 2012; Tho et al., 2016)—will be employed to convolute with the baseline S-curve. For clarification, the resultant S-curve command of the shaping method is termed S-curve–Input Shaped (SC–IS). Illustrations of the embedding and the shaping methods are shown in Fig. 1. In the SC–VSE establishment, the conditions of zero residual vibration are deficient, which leads to an underdetermined issue (see Section 3.1). This matter is resolved together with the minimum transferring time problem by formulating constrained discrete nonlinear programs (see Sections 3.2–3.3). The main challenge is to obtain a global optimal solution of those optimization problems in closed form, with which the computational effort is low, and thus online command generation can be realized. In addition, the design of the baseline S-curve command for the SC–IS scheme—to fulfill both minimum-time and actuator limits requirements—is not a trivial task because the convolution process may change the shape as well as the magnitude of the baseline S-curve. Thus, the main difficulty needing to be solved in the minimum-time SC–IS case is to enumerate all feasible shaped families, which is the crucial step to establish a sequence of optimization problems (see Section 4.1).

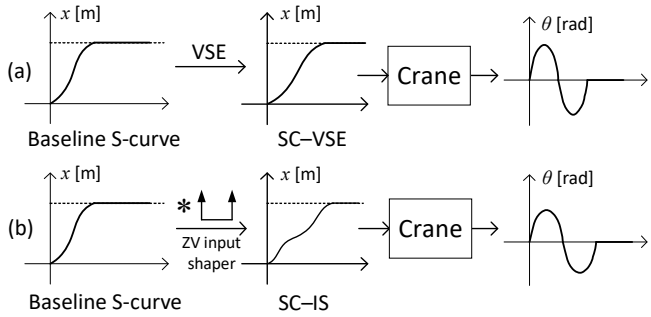


Figure 1: Description of the embedding (a) and the shaping (b) methods. The operator $*$ denotes the convolution between two signals.

To sum up, in comparison with previous studies, the contributions of this paper are as follows:

1. The minimum-time solution of the class of bang-off-bang acceleration inputs subject to both maximum velocity and maximum acceleration constraints is resolved, which fills a gap in the literature. Thus, compared with the related studies in Sun et al. (2012) and Hoang et al. (2014), the proposed ZV reference trajectory is faster.
2. The proposed minimum-time ZV motion profile can be easily computed online, which is a strong advantage compared to universal minimum-time solutions (e.g. in Auernig and Troger (1987); Kolar et al. (2017)).
3. When infinite actuator limits are allowed, this paper shows that the SC–VSE is only *two* times slower than a discontinuous command formed by a step function and the ZV input shaper (or SC–IS in equivalence), not *four* times as suggested by Singhose et al. (2010).

2. Problem formulation

The swinging motion of the payload resulting from the trolley actuation is illustrated in Fig. 2, where $x(t)$ (m) and $\theta(t)$ (rad) are the trolley position and swing angle respectively. The net force driving the trolley is denoted as $F(t)$. Moreover, m (kg), M (kg), l (m), g ($= 9.81$ m/s^2), respectively, represent the payload mass, trolley mass, rope length, and gravitational acceleration. The cable is assumed to be a massless rigid rod (i.e. all points of the rope lie on a straight line in the OXY plane), the payload is considered as a point mass, and the air resistance could be negligible. By utilizing the Euler–Lagrange method, the dynamic of the overhead crane system is described by

$$(M+m)\,\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta = F \tag{1}$$

$$l\ddot{\theta} + g\sin\theta = \ddot{x}\cos\theta. \tag{2}$$

From (1) and (2), it is recognized that one can either use $F(t)$ or $\ddot{x}(t)$ as the control input. For the motion planning task, $\ddot{x}(t)$ is the more appropriate choice for the following reasons: i) the problematic frictional forces presented in $F(t)$ can be avoided, ii) the first and second integration of $\ddot{x}(t)$ will result in the velocity $\dot{x}(t)$ and the position $x(t)$ commands, which can be easily realized by ubiquitous servo drives, and iii) the system dynamic is only dependent on the rope length l, which can be easily measured in practice. Since the control input is chosen to be $\ddot{x}(t)$, (2) is the only dynamical equation considered in the subsequent control design and analysis. By using the small angle approximation (Sun et al., 2012; Hoang et al., 2014; Chen et al., 2016), (2) can be linearized as

$$l\ddot{\theta}(t) + g\theta(t) = \ddot{x}(t). \tag{3}$$

From (3), the angular natural frequency ω_0 (rad/s) and the natural period T (s) of the overhead crane system are given by $\omega_0 = \sqrt{g/l}$ and $T = 2\pi/\omega_0$. Note that the model (3) will be employed to design the minimum-time SC–VSE and the minimum-time SC–IS in Section 3 and Section 4 respectively. Throughout the paper, it is assumed that the payload starts from rest, namely $x(0) = \dot{x}(0) = \theta(0) = \dot{\theta}(0) = 0$ (Sun et al., 2012; Hoang et al., 2014; Uchiyama et al., 2013; Béarée, 2014). Moreover, to realize a vibration-free transportation, the following terminal conditions must be fulfilled:

$$x(t_f) = x_r,\ \dot{x}(t_f) = 0,\ \theta(t_f) = 0,\ \dot{\theta}(t_f) = 0. \tag{4}$$

In (4), x_r (m) is the reference value of the trolley position. Without loss of generality, x_r is a positive constant, namely $x_r > 0$. The terminal time t_f will be minimized to enhance the transportation productivity (see Section 3 and Section 4).

In order to ensure practical feasibility, the following maximum velocity and maximum acceleration of the trolley actuator must be strictly kept:

$$|\dot{x}(t)| \le v_{\max}, \quad |\ddot{x}(t)| \le a_{\max}, \tag{5}$$

where $v_{\max} > 0$ and $a_{\max} > 0$. In the following section, the minimum-time SC–VSE will be formulated.

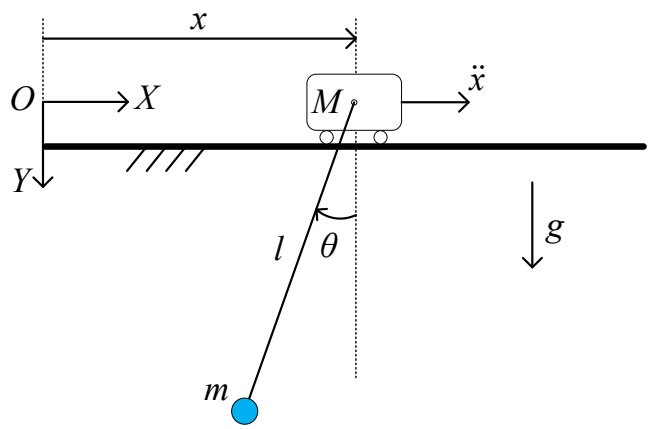


Figure 2: Mathematical model of the overhead crane system.

3. Design of the minimum-time SC–VSE

In this section, by embedding the conditions in (4) for vibration-free transportation into the baseline S-curve, the SC–VSE can be formed. It will be later shown that the no-residual-swing conditions $\theta(t_f) = \dot{\theta}(t_f) = 0$ are deficient, which leads to an underdetermined issue. This can be solved together with the minimum-time problem by establishing constrained discrete nonlinear programs. Simulation results are provided for various maneuvers with different values of $v_{\max}$ and $a_{\max}$ to verify the proposed minimum-time solution.

The position baseline S-curve trajectory is generated by integrating twice the bang-off-bang acceleration profile as depicted in Fig. 3. This piece-wise continuous function can

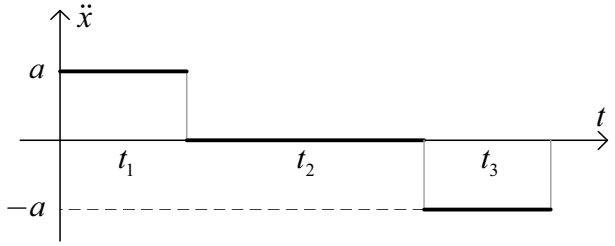


Figure 3: Parameterized bang-off-bang acceleration profile of the trolley.

be expressed by[1]

$$\ddot{x}(t) = \begin{cases} a, & 0 \le t < t_1 \\ 0, & t_1 \le t < t_1 + t_2 \\ -a, & t_1 + t_2 \le t \le t_f \ (:= t_1 + t_2 + t_3)\,. \end{cases} \tag{6}$$

In (6), all of the parameters a, t_1, t_2, and t_3 are unknowns, whose pre-conditions are given as $0 < a \le a_{\max}$, $t_1 > 0$, $t_2 \ge 0$, and $t_3 > 0$. Furthermore, it is easy to see that the baseline S-curve obtained from (6) reaches its highest velocity ($= at_1$) at $t = t_1$. Therefore, the maximum velocity constraint $|\dot{x}(t)| \le v_{\max}$ becomes $at_1 \le v_{\max}$. The following subsection will establish additional constraints for these unknown variables to actively gain a vibration-free transferring process.

3.1. Embedding vibration-free transportation conditions

By exploiting the vibration-free transportation condition (4), essential constraints on a, t_1, t_2, and t_3 can be further obtained. First, response of the system (3) under the action of the control input (6) will be derived.

During the acceleration interval, namely $0 \le t < t_1$, by substituting $\ddot{x}(t) = a$ into (3) and solving the second-order differential equation with a constant excitation, the following result is obtained:

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left(1 - \cos\omega_0 t\right), & \dot{\theta}(t) &= \frac{a\omega_0}{g}\sin\omega_0 t \\ x(t) &= 0.5at^2, & \dot{x}(t) &= at. \end{aligned} \tag{7}$$

Similarly, when $t_1 \le t < t_1 + t_2$ (i.e. uniform interval), $\ddot{x}(t) = 0$ holds. Therefore

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left[\cos\left(\omega_0\left(t - t_1\right)\right) - \cos\omega_0 t\right] \\ \dot{\theta}(t) &= -\frac{a\omega_0}{g}\left[\sin\left(\omega_0\left(t - t_1\right)\right) - \sin\omega_0 t\right] \\ x(t) &= at_1 t - 0.5at_1^2 \\ \dot{x}(t) &= at_1. \end{aligned} \tag{8}$$

Finally, when $t_1 + t_2 \le t \le t_f$ (i.e. deceleration interval), $\ddot{x}(t) = -a$. Hence

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left[\sum_{i=1}^{2}\cos\left(\omega_0\left(t - \sum_{j=1}^{i} t_j\right)\right) - \cos\omega_0 t - 1\right] \\ \dot{\theta}(t) &= \frac{-a\omega_0}{g}\left[\sum_{i=1}^{2}\sin\left(\omega_0\left(t - \sum_{j=1}^{i} t_j\right)\right) - \sin\omega_0 t\right] \\ x(t) &= -\frac{1}{2}at^2 + a\left(2t_1 + t_2\right)t - a\left(t_1^2 + t_1 t_2 + \frac{1}{2}t_2^2\right) \\ \dot{x}(t) &= a\left(-t + 2t_1 + t_2\right). \end{aligned} \tag{9}$$

From (9), the zero velocity condition of the trolley at t_f—$\dot{x}\left(t_f\right) = 0$—gives $a\left(-t_f + 2t_1 + t_2\right) = 0$, which can be easily developed to $t_f - 2t_1 - t_2 = 0$ since $a > 0$. Note that $t_f = t_1 + t_2 + t_3$; thus, $t_1 = t_3$. This means that the duration of the acceleration and deceleration periods must be similar. Using this result, the condition on the target trolley position, namely $x\left(t_f\right) = x_r$, can be exploited as

$$at_1\left(t_1 + t_2\right) = x_r. \tag{10}$$

The condition of zero swing angle at t_f can be further developed by setting $\theta\left(t_f\right) = 0$ (in (9)). Consequently, it yields

$$\cos\left(\omega_0\left(t_1 + t_2\right)\right) + \cos\omega_0 t_1 = 1 + \cos\left(\omega_0\left(2t_1 + t_2\right)\right). \tag{11}$$

Using the cosine sum-to-product trigonometric identity $\cos p + \cos q = 2\cos\left(\left(p + q\right)/2\right)\cos\left(\left(p - q\right)/2\right)$ on the left-hand-side (LHS) and $\cos 2p = 2\cos^2 p - 1$ on the right-hand-side (RHS) of (11) results in

$$\cos\left(\frac{\omega_0\left(2t_1 + t_2\right)}{2}\right)\left[\cos\left(\frac{\omega_0 t_2}{2}\right) - \cos\left(\frac{\omega_0\left(2t_1 + t_2\right)}{2}\right)\right] = 0. \tag{12}$$

Let $\mathbb{N}^+$ represent the set of positive integers. The solution of (12) can be found as

$$\left[\begin{aligned} t_1 &= k_1 T \\ t_1 + t_2 &= k_1 T \\ 2t_1 + t_2 &= \left(k_1 - 0.5\right)T \end{aligned}\right., \quad k_1 \in \mathbb{N}^+. \tag{13}$$

Lastly, to ensure the zero swing rate condition $\dot{\theta}\left(t_f\right) = 0$, from (9), it needs

$$\sin\left(\omega_0\left(t_1 + t_2\right)\right) + \sin\omega_0 t_1 = \sin\left(\omega_0\left(2t_1 + t_2\right)\right). \tag{14}$$

The solution of (14) can be similarly obtained as follows:

$$\left[\begin{aligned} t_1 &= k_2 T \\ t_1 + t_2 &= k_2 T \\ 2t_1 + t_2 &= k_2 T \end{aligned}\right., \quad k_2 \in \mathbb{N}^+. \tag{15}$$

In sum, the zero residual vibration conditions $\theta\left(t_f\right) = \dot{\theta}\left(t_f\right) = 0$ can be obtained by subsuming (13) and (15).

[1] Although the acceleration profile of the motion is only piece-wise continuous, the resultant position S-curve command is continuously differentiable. Thus, the motion's smoothness is ensured.

By using the set associative, commutative, and distributive laws (Rudin, 1976), the subsumption of (13) and (15) leads to either of the following cases:

$$t_1 = k_1 T,\ k_1 \in \mathbb{N}^+ \tag{16}$$

$$\text{or } t_1 + t_2 = k_2 T,\ k_2 \in \mathbb{N}^+. \tag{17}$$

According to the above analysis, two equations $\theta(t_f) = 0$ and $\dot{\theta}(t_f) = 0$ are reduced to a single equation—either (16) or (17). Therefore, there are now only two equations ((10)&(16) or (10)&(17)) to solve for the three unknowns a, t_1, and t_2, which is clearly an underdetermined problem. Furthermore, the actuator limits represented by $v_{\max}$ and $a_{\max}$ must be taken into account. In order to resolve the above-mentioned underdetermined issue, the common strategy in the literature is to fix a specific quantity and utilize the constraints $|\dot{x}(t)| \le v_{\max}$ and $|\ddot{x}(t)| \le a_{\max}$ to determine the value of a. For instance, Sun et al. (2012) set k_1 (in (16)) to $k_1 = 1$, namely $t_1 = T$. Then, a is calculated by $a = v/T$ where the velocity of the uniform interval v is determined by $v = \min\{v_{\max},\ Ta_{\max},\ x_r/T\}$. In contrast, Hoang et al. (2014) fixed $v = v_{\max}$ and varied the index k_1 until the maximum acceleration constraint is fulfilled. These solutions, however, do not guarantee a time-optimal choice (see comparative results in Section 3.5). Therefore, this paper will establish constrained discrete nonlinear programs to solve both underdetermined and time-optimal problems while ensuring that the actuator limits are complied with. The results of these optimization problems are admitted in closed form, and thus no intensive computation is required. The following subsection will discuss the optimal solution for the first case (i.e. $t_1 = k_1 T,\ k_1 \in \mathbb{N}^+$).

3.2. Case I of SC–VSE: $t_1 = k_1 T,\ k_1 \in \mathbb{N}^+$

To begin, the essential equality and inequality constraints needing to be taken into account are summarized as follows: 1) trajectory constraints: $t_1 > 0$ and $t_2 \ge 0$, 2) actuator limits: $0 < a \le a_{\max}$ and $at_1 \le v_{\max}$, and 3) vibration-free transportation conditions: $t_1 = k_1 T$, $k_1 \in \mathbb{N}^+$ and $at_1(t_1 + t_2) = x_r$ (see Section 3.1). By considering all of the above constraints and taking the total transferring time t_f as the cost function, the following optimization problem can be established:

$$\begin{aligned} \min_{a,t_1,t_2} \quad & t_f = 2t_1 + t_2 \\ \text{s.t.} \quad & at_1(t_1 + t_2) = x_r \\ & t_1 = k_1 T\ (k_1 \in \mathbb{N}^+),\ t_2 \ge 0 \\ & 0 < a \le a_{\max},\ at_1 \le v_{\max}. \end{aligned} \tag{18}$$

Note that (18) is a discrete nonlinear program since t_1 can only be a multiple of the natural period T, whereas a and t_2 are continuous. Eliminating the two variables t_1 and t_2 using two equalities, (18) becomes the following mixed nonlinear–integer program:

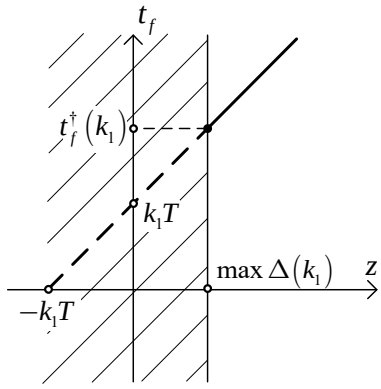


Figure 4: Optimal solution of the linear optimization problem (21).

$$\begin{aligned} \min_{a,k_1} \quad & t_f = k_1 T + \frac{x_r}{ak_1 T} \\ \text{s.t.} \quad & k_1 \in \mathbb{N}^+ \\ & 0 < a \le a_{\max},\ a \le \frac{x_r}{(k_1 T)^2},\ a \le \frac{v_{\max}}{k_1 T}. \end{aligned} \tag{19}$$

In (19), the inequality constraint $a \le x_r/(k_1 T)^2$ is a result of the condition $t_2 \ge 0$ in (18) applied to $t_2 = x_r/(ak_1 T) - k_1 T$. For a fixed k_1, (19) is actually a quasi-linear program, i.e., the original nonlinear optimization problem (19) can be converted to a linear one through a suitable variable substitution (see the proof of Lemma 1). In order to solve (19), a two-step procedure is proposed. Denote a set $\Delta(k_1)$ as below

$$\Delta(k_1) = \left\{ k_1 T,\ \frac{x_r}{k_1 T a_{\max}},\ \frac{x_r}{v_{\max}} \right\}. \tag{20}$$

In the first step, the optimal solution of (19) is established for each $k_1 \in \mathbb{N}^+$ by the following lemma:

Lemma 1. *Consider the mixed nonlinear–integer optimization problem (19). For a fixed k_1 ($k_1 \in \mathbb{N}^+$), the minimizer $a^\dagger(k_1)$ and the optimum cost function $t_f^\dagger(k_1)$ of (19) are given by*

$$a^\dagger(k_1) = \frac{x_r}{k_1 T \max\Delta(k_1)} \text{ and } t_f^\dagger(k_1) = k_1 T + \max\Delta(k_1).$$

Proof. By assigning $z = x_r/(ak_1 T)$, the original nonlinear program (19) can be transformed to a linear optimization problem (21), where $\mathbb{R}^+$ is the set of positive real numbers.

$$\begin{aligned} \min_{z \in \mathbb{R}^+} \quad & t_f = k_1 T + z \\ \text{s.t.} \quad & z \ge k_1 T,\ z \ge \frac{x_r}{k_1 T a_{\max}},\ z \ge \frac{x_r}{v_{\max}}. \end{aligned} \tag{21}$$

It can be seen that the feasible region of (21) is $z \ge \max\Delta(k_1)$. Therefore, based on Fig. 4, the minimizer of (21) is $z^\dagger = \max\Delta(k_1)$ and $t_f^\dagger = k_1 T + z^\dagger$. Using the relation $z = x_r/(ak_1 T)$ leads to the results $a^\dagger(k_1) = x_r/(k_1 T \max\Delta(k_1))$ and $t_f^\dagger(k_1) = k_1 T + \max\Delta(k_1)$. This completes the proof. □

In fact, through Lemma 1, the first step will provide all local minima of the optimization problem (19), which are infinite in number. In the second step, these local minima will be compared to each other to determine the global optimum one. The result of the second step is summarized in Theorem 1.

Theorem 1. *The global minimizer $k_1^\star$ and $a^\star$ of the mixed nonlinear–integer program (19) are*

$$k_1^\star = \begin{cases} 1 & \text{if } \bar{k}_1 = 1 \\ \bar{k}_1 & \text{if } \bar{k}_1 > 1 \ \& \ \mu < 0 \\ \bar{k}_1 - 1 & \text{if } \bar{k}_1 > 1 \ \& \ \mu \geq 0 \end{cases} ; \ a^\star = \frac{x_r}{k_1^\star T \max \Delta\left(k_1^\star\right)},$$

in which $\mu = T + \max\Delta\left(\bar{k}_1\right) - \max\Delta\left(\bar{k}_1 - 1\right)$. The pivotal index is $\bar{k}_1 = 1$ if $\max\Delta(1) \neq x_r/\left(Ta_{\max}\right)$. Otherwise, namely $\max\Delta(1) = x_r/\left(Ta_{\max}\right)$, $\bar{k}_1$ is given by

$$\bar{k}_1 = \max\left\{k_1 \in \mathbb{N}^+ : \max\Delta\left(k_1\right) = \frac{x_r}{k_1 T a_{\max}}\right\} + 1.$$

Moreover, the global minimum cost function is

$$t_f^\star\left(k_1^\star\right) = k_1^\star T + \max\Delta\left(k_1^\star\right).$$

The proof of Theorem 1 is given in Appendix A. The computation of the global optimal solution (for Case I) based on Theorem 1 should begin from the determination of the pivotal index $\bar{k}_1$, and then μ can be computed. From $\bar{k}_1$ and μ, the global optimum index $k_1^\star$ can be determined. Finally, the optimal acceleration $a^\star$ and the minimum transferring time $t_f^\star\left(k_1^\star\right)$ can be calculated. The optimal values of t_1 and t_2 are computed by $t_1^\star = k_1^\star T$ and $t_2^\star = t_f^\star - 2t_1^\star$ respectively. Now the global minimum-time solution of SC–VSE in Case I has been established. Next, Case II ($t_1 + t_2 = k_2 T$, $k_2 \in \mathbb{N}^+$) will be investigated.

Remark 1. In the first step of solving the mixed nonlinear–integer program (19), it was chosen to find the local minimum for each $k_1 \in \mathbb{N}^+$ rather than for each a. If the latter strategy is employed, the optimization problem needing to be solved in the second step becomes much more complex. Therefore, fixing k_1 first is the more reasonable approach.

3.3. Case II of SC–VSE: $t_1 + t_2 = k_2 T$, $k_2 \in \mathbb{N}^+$

In order to draw a distinction with Case I, in Case II, the unknown parameters a, t_1, t_2, and t_f are denoted as $\tilde{a}$, $\tilde{t}_1$, $\tilde{t}_2$, and $\tilde{t}_f$ respectively. Similar to Case I, by gathering all essential trajectory, actuator limits, and vibration-free transportation constraints and taking the total transferring time as the cost function, the following discrete nonlinear program can be established:

$$\begin{aligned} \min_{\tilde{a},\tilde{t}_1,\tilde{t}_2} \quad & \tilde{t}_f = 2\tilde{t}_1 + \tilde{t}_2 \\ \text{s.t.} \quad & \tilde{a}\tilde{t}_1\left(\tilde{t}_1 + \tilde{t}_2\right) = x_r \\ & \tilde{t}_1 + \tilde{t}_2 = k_2 T \ (k_2 \in \mathbb{N}^+),\ \tilde{t}_1 > 0,\ \tilde{t}_2 \geq 0 \\ & 0 < \tilde{a} \leq a_{\max},\ \tilde{a}\tilde{t}_1 \leq v_{\max}. \end{aligned} \tag{22}$$

By using the equality constraints, the two variables $\tilde{t}_1$ and $\tilde{t}_2$ can be eliminated, and thus (22) can be transformed into the following mixed nonlinear–integer program:

$$\begin{aligned} \min_{\tilde{a},k_2} \quad & \tilde{t}_f = k_2 T + \frac{x_r}{\tilde{a} k_2 T} \\ \text{s.t.} \quad & k_2 \geq \frac{x_r}{T v_{\max}},\ k_2 \geq \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}},\ k_2 \in \mathbb{N}^+ \\ & \frac{x_r}{\left(k_2 T\right)^2} \leq \tilde{a} \leq a_{\max}. \end{aligned} \tag{23}$$

By conducting an analogous procedure to Case I, the solution of the optimization problem (23) is stated in Theorem 2. To begin, a new set $\beta\left(k_2\right)$ is denoted as follows:

$$\beta\left(k_2\right) = \left\{k_2,\ \frac{x_r}{T v_{\max}},\ \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}}\right\}. \tag{24}$$

Theorem 2. *The global optimal solution of the mixed nonlinear–integer optimization problem (23) is*

$$\begin{aligned} k_2^\star &= \min\left\{k_2 \in \mathbb{N}^+ : \max\beta\left(k_2\right) = k_2\right\} \\ \tilde{a}^\star &= a_{\max}. \end{aligned}$$

Moreover, the global minimum cost function is given by

$$\tilde{t}_f^\star\left(k_2^\star\right) = k_2^\star T + \frac{x_r}{k_2^\star T a_{\max}}.$$

The proof of Theorem 2 can be found in Appendix B. By using the result of Theorem 2, the global optimal solution of SC–VSE in Case II can be computed. Note that, the optimum values of $\tilde{t}_1$ and $\tilde{t}_2$ are determined by $\tilde{t}_1^\star = \tilde{t}_f^\star - k_2^\star T$ and $\tilde{t}_2^\star = 2k_2^\star T - \tilde{t}_f^\star$ respectively.

3.4. Comparison between Case I and Case II

Although Case I ($t_1 = k_1 T$) and Case II ($t_1 + t_2 = k_2 T$) will both result in a vibration-free transferring process, they exhibit some fundamentally distinct properties, which will be clarified in the following discussion. In Case I, by substituting $t_1 = k_1 T$ ($k_1 \in \mathbb{N}^+$) into (8), it yields

$$\theta\left(t\right) = \dot{\theta}\left(t\right) = 0,\ \forall t \in \left[t_1, t_1 + t_2\right). \tag{25}$$

The condition of (25) implies that, in Case I, there is never any residual vibration during the uniform interval (i.e. $t_1 \leq t < t_1 + t_2$). In contrast, no similar property is present in Case II except when the resultant $\tilde{t}_f^\star$ is also a multiple of the natural period T.

The second different property between Case I and Case II can be revealed by letting $v_{\max} \to +\infty$ and $a_{\max} \to +\infty$, namely, when infinite maximum velocity and maximum acceleration are allowed. In such a context, for Case I, the following can be obtained:

$$\lim_{\lambda \to +\infty} \max\left\{T,\ \frac{x_r}{T a_{\max}},\ \frac{x_r}{v_{\max}}\right\} = T, \tag{26}$$

where λ is defined by $\lambda = \min\{v_{\max}, a_{\max}\}$. Therefore, $\lim_{\lambda\to+\infty} \max \Delta(1) = T$. Consequently, according to Theorem 1, $k_1^\star = 1$ as $\lambda \to +\infty$ and hence

$$\lim_{\lambda\to+\infty} t_f^\star(k_1^\star) = T + T = 2T. \tag{27}$$

Similarly, for Case II, it can be shown that $\lim_{\lambda\to+\infty} \max \beta(k_2) = k_2$. Therefore, $k_2^\star = 1$ as $\lambda \to +\infty$. As a result, based on Theorem 2

$$\lim_{\lambda\to+\infty} \tilde{t}_f^\star(k_2^\star) = T. \tag{28}$$

The results of (27) and (28) can be interpreted as follows: with infinite velocity and acceleration limits, the minimum transfer time for Case I is twice the natural period T, whereas it only requires one natural period for Case II. In other words, it is not possible to obtain a transfer time less than one natural period in the case of SC–VSE.

The vibration-free transportation using SC–VSE has two candidates: Case I and Case II. Therefore, in practice, the SC–VSE scheme will choose the one which produces a smaller optimal transferring time. For instance, as $v_{\max} \to +\infty$ and $a_{\max} \to +\infty$, it holds that $t_f^\star(\to 2T) > \tilde{t}_f^\star(\to T)$, and thus Case II should be selected. Note that, Singhose et al. (2010) stated that the position S-curve (including both triangular and trapezoidal velocity profiles) must be *four* times longer than a command formed by shaping a step function with a ZV input shaper. The previous discussion showed that the above conclusion is not generally true, namely that it is only correct for Case I, but not for Case II. This can be explained as follows. When a step position command is employed as in Singhose et al. (2010), it automatically implies that infinite velocity and acceleration can be admitted. The shaping process between a step position command and the ZV input shaper therefore introduces a total transferring time of $T/2$. Under the same conditions, Case II of SC–VSE can be selected, namely that a vibration-free transportation can be guaranteed with a total time of T (see (28)), which is only *two* times longer. The shortcoming of Singhose et al. (2010) stems from the fact that Case II is not fully considered in the analysis/design. In order to clearly verify the above argument, the minimum-time SC–VSE is computed for the following configuration: $l = 1.4$ m ($T = 2.37$ s), $x_r = 1$ m, $v_{\max} = 5$ m/s, and $a_{\max} = 5$ m/s^2 (high maximum velocity and maximum acceleration are allowed). By applying the results from Section 3.2 and Section 3.3, Case II is finally selected. The optimal solution is $\tilde{t}_1^\star = 0.084$ s, $\tilde{t}_2^\star = 2.289$ s, $\tilde{a}^\star = 5$ m/s^2. The minimum transferring time of SC–VSE is $\tilde{t}_f^\star = 2.458$ s, which is only 2.071 times longer than $T/2$ (less than four times).

3.5. Simulation results of the minimum-time SC–VSE

In this section, simulation results of the minimum-time SC–VSE and comparisons with the related studies in Sun et al. (2012) and Hoang et al. (2014) are given. In order to verify the established theoretical results, the simulation study is conducted under various conditions of the actuator limits, i.e., both maximum velocity $v_{\max}$ and maximum acceleration $a_{\max}$ will be varied. The transferring distance is 2 meters ($x_r = 2$ m). Moreover, necessary parameters for all testcases are shown in Table 1. The computations are conducted on a personal computer with 2.7-GHz Intel Core i5-4310M and 8 GB of RAM. Computational results of the minimum-time SC–VSE are aggregated in Table 2. On average, it takes 0.013 second to compute the minimum-time SC–VSE for each testcase showing that online computation is feasible.

The simulation results of all testcases are illustrated in Figs. 5–7. It is recognized that the larger the actuator limits, the smaller the total transferring time would be (with an increment of the peak value of swing angle). In all testcases, the trolley precisely arrives at the desired destination with no vibration after the transfer. Moreover, the maximum velocity and maximum acceleration constraints

Table 1: Configuration of testcases (TCs).

TC#	$v_{\max}$	$a_{\max}$	Rope length	Nat. period
A1	0.25 m/s	0.05 m/s^2	$l = 1.4$ m	$T = 2.37$ s
A2	0.25 m/s	0.1 m/s^2	$l = 1$ m	$T = 2.01$ s
A3	0.5 m/s	0.5 m/s^2	$l = 1.4$ m	$T = 2.37$ s

Table 2: Computational results of the minimum-time SC–VSE.

TC#	Selection	$a^\star \vert \tilde{a}^\star$	$t_1^\star \vert \tilde{t}_1^\star$	$t_2^\star \vert \tilde{t}_2^\star$	$k_1^\star \vert k_2^\star$	$t_f^\star \vert \tilde{t}_f^\star$
A1	Case I	0.05 m/s^2	4.75 s	3.68 s	$k_1^\star = 2$	13.17 s
A2	Case II	0.1 m/s^2	2.49 s	5.53 s	$k_2^\star = 4$	10.52 s
A3	Case II	0.5 m/s^2	0.84 s	3.90 s	$k_2^\star = 2$	5.59 s

$x_r = 2$ m. Case I: Section 3.2, Case II: Section 3.3.

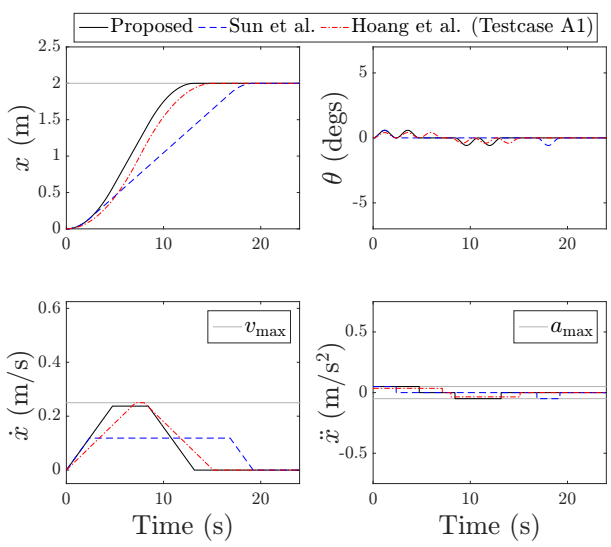


Figure 5: Simulation result of the minimum time SC–VSE and its comparison with related studies (testcase A1).

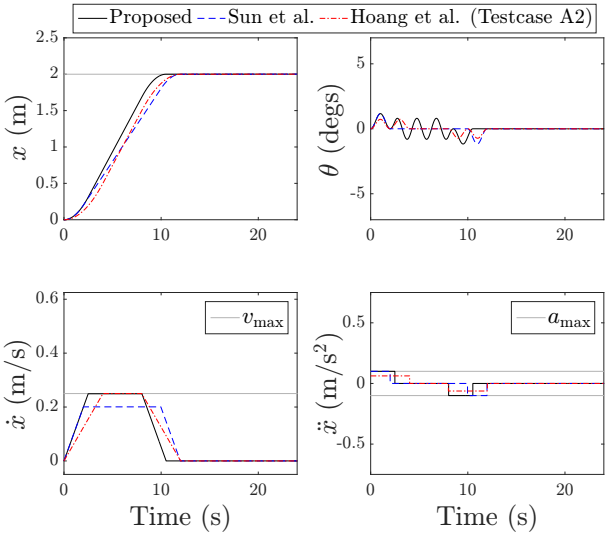


Figure 6: Simulation result of the minimum time SC–VSE and its comparison with related studies (testcase A2).

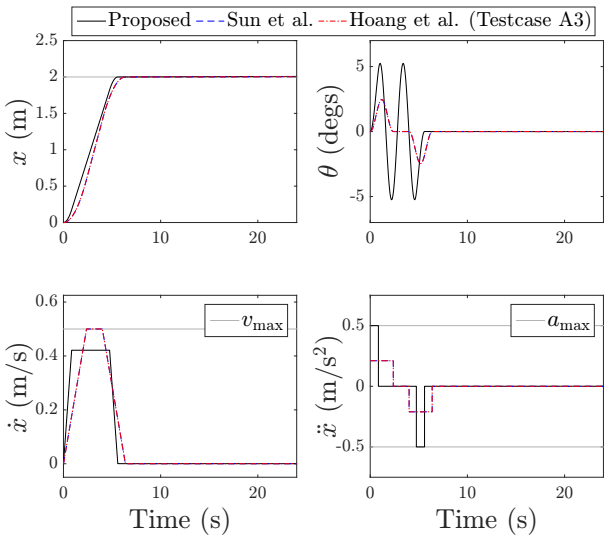


Figure 7: Simulation result of the minimum time SC–VSE and its comparison with related studies (testcase A3).

are strictly complied with. Therefore, the effectiveness of the minimum-time SC–VSE scheme is confirmed.

The proposed minimum-time SC–VSE is now compared to the related studies (Sun et al., 2012; Hoang et al., 2014) in the same bang-off-bang acceleration profile family. Note that, to draw a fair comparison, the constraint of maximum allowable swing angle presented in Sun et al. (2012) and Hoang et al. (2014) are relaxed (i.e. setting $\theta_{\text{ub}} = +\infty$ in Sun et al. (2012) and $\theta_{\max} = +\infty$ in Hoang et al. (2014)). The comparative results are shown in Table 3 and Figs. 5–7. It can be seen that the proposed minimum-time SC–VSE provides considerably smaller transfer time than the existing studies. Specifically, it is 6.06 s and 1.95 s faster in testcase A1, 1.46 s and 1.49 s faster in testcase A2, and 0.78 s and 0.78 s faster in testcase A3, in

Table 3: Comparison of the min-time SC–VSE with related studies.

TC#	Transferring time ($x_r = 2$ m)		
	Proposed	Sun et al. (2012)	Hoang et al. (2014)
A1	**13.17 s**	19.23 s	15.12 s
A2	**10.52 s**	11.98 s	12.01 s
A3	**5.59 s**	6.37 s	6.37 s

comparison with Sun et al. (2012) and Hoang et al. (2014) respectively. Therefore, the proposed minimum-time SC–VSE demonstrated its advantage in terms of transfer time.

4. Design of the minimum-time SC–IS

4.1. Formulation

In the SC–IS scheme, to gain a vibration-free transportation, the baseline S-curve is modulated by convolving with the well-known ZV input shaper, which is given by (Pereira et al., 2012)

$$I(t) = 0.5\left[\delta(t) + \delta\left(t - T/2\right)\right], \tag{29}$$

where $\delta(.)$ is the Dirac's delta generalized function. The position baseline S-curve is still generated by integrating twice the bang-off-bang acceleration profile $\bar{R}(t)$, which takes a similar form of (6), that is

$$\bar{R}(t) = \begin{cases} \bar{a}, & 0 \le t < \bar{t}_1 \\ 0, & \bar{t}_1 \le t < \bar{t}_1 + \bar{t}_2 \\ -\bar{a}, & \bar{t}_1 + \bar{t}_2 \le t \le \bar{t}_f \ \ (:= 2\bar{t}_1 + \bar{t}_2)\,. \end{cases} \tag{30}$$

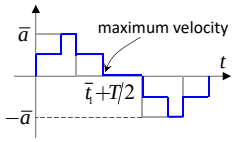
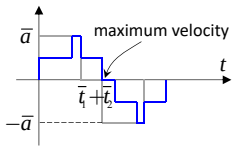
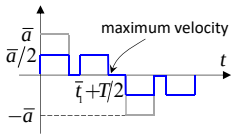
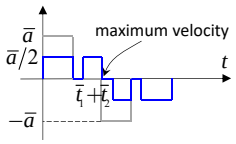
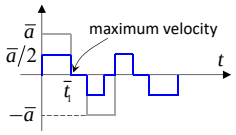
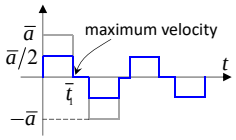
The convolution product between (30) and the ZV input shaper will result in a shaped acceleration command, whose double integration is the SC–IS. As a matter of fact, the convolution process will possibly change the shape as well as the magnitude of the original command. Therefore, the present task is to design a baseline function in the form of (30), which ensures the resultant shaped command: 1) achieves a minimum transferring time and 2) respects the maximum allowable velocity and acceleration. This design task for the SC–IS scheme is not trivial.

If $\bar{Q}(t)$ is denoted as the shaped acceleration function in time domain, it can be computed by

$$\bar{Q}(t) = \bar{R}(t) * I(t), \tag{31}$$

where $I(t)$ and $\bar{R}(t)$ are given in (29) and (30) respectively. The operator $*$ denotes the convolution between two signals. In the case of SC–IS scheme, the actual acceleration input feeding to the crane system in (3) is $\ddot{\bar{x}}(t) = \bar{Q}(t)$, where $\ddot{x}(t)$ is replaced by $\ddot{\bar{x}}(t)$ to draw a distinction with the SC–VSE technique. To compute $\bar{Q}(t)$ in an explicit form, the Laplace transformation method is

Table 4: All possibilities of the shaped command $\bar{Q}(t)$ and optimization configurations of the minimum-time SC–IS.

P#	$\bar{Q}(t)$	Optimization Configuration				
		Condition	Acc. Constraint	Vel. Constraint	Equality Constraint	Cost Function
P1		$\bar{t}_1 \geq T/2$ $\bar{t}_2 \geq T/2$	$0 < \bar{a} \leq a_{\max}$	$\bar{a}\bar{t}_1 \leq v_{\max}$		
P2		$\bar{t}_1 \geq T/2$ $0 \leq \bar{t}_2 \leq T/2$	$0 < \bar{a} \leq a_{\max}$	$\frac{\bar{a}}{2}\left(2\bar{t}_1 + \bar{t}_2 - \frac{T}{2}\right) \leq v_{\max}$		
P3		$0 < \bar{t}_1 \leq T/2$ $\bar{t}_1 + \bar{t}_2 \geq T/2$ $\bar{t}_2 \geq T/2$	$0 < \bar{a} \leq 2a_{\max}$	$\bar{a}\bar{t}_1 \leq v_{\max}$		
P4		$0 < \bar{t}_1 \leq T/2$ $\bar{t}_1 + \bar{t}_2 \geq T/2$ $0 \leq \bar{t}_2 \leq T/2$	$0 < \bar{a} \leq 2a_{\max}$	$\frac{\bar{a}}{2}\left(2\bar{t}_1 + \bar{t}_2 - \frac{T}{2}\right) \leq v_{\max}$	$\bar{a}\bar{t}_1\left(\bar{t}_1 + \bar{t}_2\right) = x_r$	$\bar{J} = 2\bar{t}_1 + \bar{t}_2 + \frac{T}{2}$
P5		$0 < \bar{t}_1 \leq T/2$ $\bar{t}_1 + \bar{t}_2 \leq T/2$ $2\bar{t}_1 + \bar{t}_2 \geq T/2$	$0 < \bar{a} \leq 2a_{\max}$	$\bar{a}\bar{t}_1 \leq 2v_{\max}$		
P6		$0 < \bar{t}_1 \leq T/2$ $\bar{t}_1 + \bar{t}_2 \leq T/2$ $2\bar{t}_1 + \bar{t}_2 \leq T/2$	$0 < \bar{a} \leq 2a_{\max}$	$\bar{a}\bar{t}_1 \leq 2v_{\max}$		

Light gray line: unshaped/baseline command $\bar{R}(t)$ (in (30)). Bold blue line: shaped command $\bar{Q}(t)$ (in (34)).

employed. The Laplace transform of the baseline function $\bar{R}(t)$ and the input shaper $I(t)$ can be found as

$$\bar{R}(s) = \frac{-\bar{a}}{s}\left(e^{-s\bar{t}_1} + e^{-s(\bar{t}_1+\bar{t}_2)} - e^{-s(2\bar{t}_1+\bar{t}_2)} - 1\right) \quad (32)$$

$$I(s) = 0.5\left(1 + e^{-\frac{T}{2}s}\right). \quad (33)$$

Performing the Laplace transform on both sides of (31) yields $\bar{Q}(s) = \bar{R}(s)I(s)$. By taking the inverse Laplace transform, $\bar{Q}(t)$ can be calculated by

$$\begin{aligned}\bar{Q}(t) = \frac{\bar{a}}{2}\,[&-H(t-\bar{t}_1) - H(t-\bar{t}_1-\bar{t}_2) - H(t-\bar{t}_1-T/2)\\ &- H(t-\bar{t}_1-\bar{t}_2-T/2) + H(t-2\bar{t}_1-\bar{t}_2)\\ &+H(t-T/2) + H(t-2\bar{t}_1-\bar{t}_2-T/2) + 1]\,,\end{aligned} \quad (34)$$

where $H(.)$ is the Heaviside function. According to (34) and the anti-symmetrical property of $\bar{Q}(t)$ along the vertical line $t = 0.5\,(\bar{t}_f + T/2)$, all possibilities of $\bar{Q}(t)$ are listed in Table 4.

The classification of shaped families P1–P6 in Table 4 depends entirely on how large $\bar{t}_1$ and $\bar{t}_2$ of the unshaped command $\bar{R}(t)$ compare with $T/2$ (half of the natural period). For instance, in the case of P1 ($\bar{t}_1 \geq T/2$ and $\bar{t}_2 \geq T/2$), the shaped command $\bar{Q}(t)$ (bold blue line) reserves the maximum magnitude $\bar{a}$ of the baseline function $\bar{R}(t)$ (light gray line). Moreover, the shaped velocity—obtained by integrating $\bar{Q}(t)$—reaches its maximum value $\bar{a}\bar{t}_1$ at $t = \bar{t}_1 + T/2$. On the other hand, in the P4 family, the maximum value of $\bar{Q}(t)$ is reduced to half compared with the original command; the peak of the shaped velocity attains at $t = \bar{t}_1 + \bar{t}_2$ with a magnitude of $0.5\bar{a}(2\bar{t}_1 + \bar{t}_2 - T/2)$. Such a classification is needed because essential inequality constraints generated from the actuator limits $v_{\max}$ and $a_{\max}$ are different for each family of shapes.

For every family P1–P6, by letting the total transferring time of the shaped command as the cost function and collecting all necessary equality and inequality constraints

Table 5: Comparative results of the minimum-time SC–IS and the minimum-time SC–VSE.

TC#	Minimum-time SC–IS ($x_r = 1$ m)					Minimum-time SC–VSE ($x_r = 1$ m)					
	Selected family	$\bar{a}^\star$	$\bar{t}_1^\star$	$\bar{t}_2^\star$	$\bar{J}^\star$	Selection	$a^\star \vert \tilde{a}^\star$	$t_1^\star \vert \tilde{t}_1^\star$	$t_2^\star \vert \tilde{t}_2^\star$	$k_1^\star \vert k_2^\star$	$t_f^\star \vert \tilde{t}_f^\star$
A1	P2	0.05 m/s^2	4.47 s	0 s	10.13 s	Case II	0.05 m/s^2	4.21 s	0.53 s	$k_2^\star = 2$	**8.96 s**
A2	P3	0.2 m/s^2	1.00 s	3.98 s	6.99 s	Case II	0.1 m/s^2	2.49 s	1.52 s	$k_2^\star = 2$	**6.50 s**
A3	P5	1 m/s^2	1.00 s	0 s	**3.19 s**	Case II	0.5 m/s^2	0.84 s	1.53 s	$k_2^\star = 1$	3.22 s

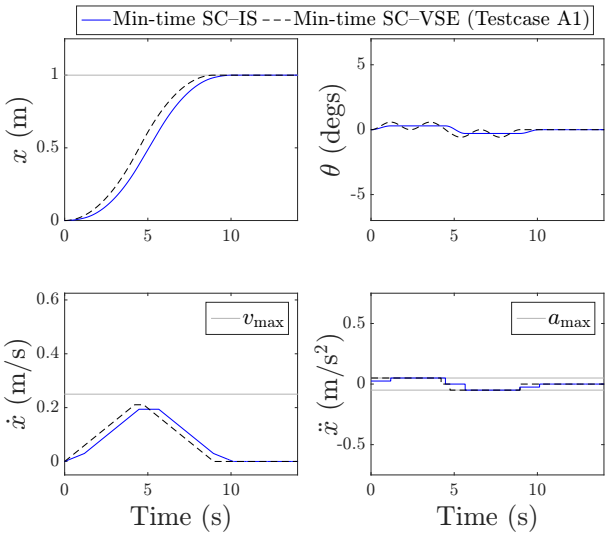


Figure 8: Simulation result of the minimum-time SC–IS and its comparison with the minimum-time SC–VSE in testcase A1.

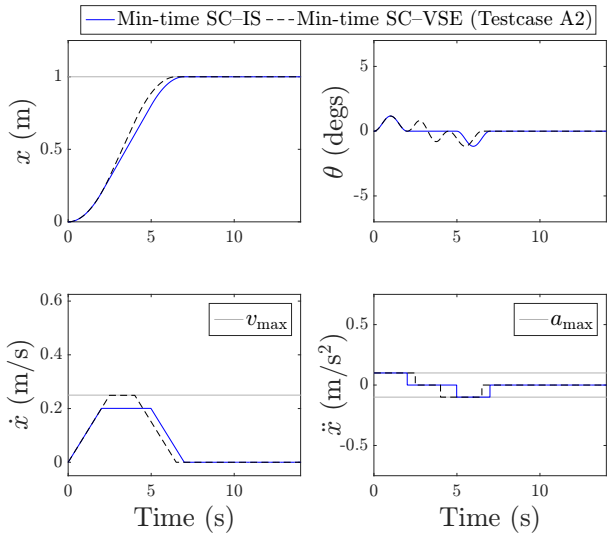


Figure 9: Simulation result of the minimum-time SC–IS and its comparison with the minimum-time SC–VSE in testcase A2.

(shown in Table 4), an optimization problem can be established to solve for unknowns $\bar{a}$, $\bar{t}_1$, and $\bar{t}_2$. Six nonlinear programs are formulated, and then sequentially solved by the *fmincon* routine of MATLAB. In fact, the optimal solutions of these nonlinear programs can also be analytically obtained by invoking the Karush-Kuhn-Tucker (KKT) optimality condition. However, the present problem has a considerably large number of families, and thus a numerical method is preferable. Finally, the (feasible) optimal solution of the family which produces the smallest transferring time will be chosen as the eventual answer for the minimum-time SC–IS scheme.

4.2. Simulation results of the minimum-time SC–IS

In an analogous manner to the SC–VSE scheme, simulation studies of the minimum-time SC–IS are conducted in three testcases with different values of $v_{\max}$ and $a_{\max}$ as shown in Table 1. In addition, to facilitate the subsequent experiments in Section 5, where the running length of the trolley is limited, the transferring distance is changed to 1 meter (i.e. $x_r = 1$ m). Computational results of the minimum-time SC–IS are summarized in Table 5, where $\bar{(.)}^\star$ denotes the resultant optimal value of $\bar{(.)}$. On average, the computational time for each testcase is 0.26 second. Therefore, similar to the minimum-time SC–VSE, the minimum-time SC–IS can also be computed online.

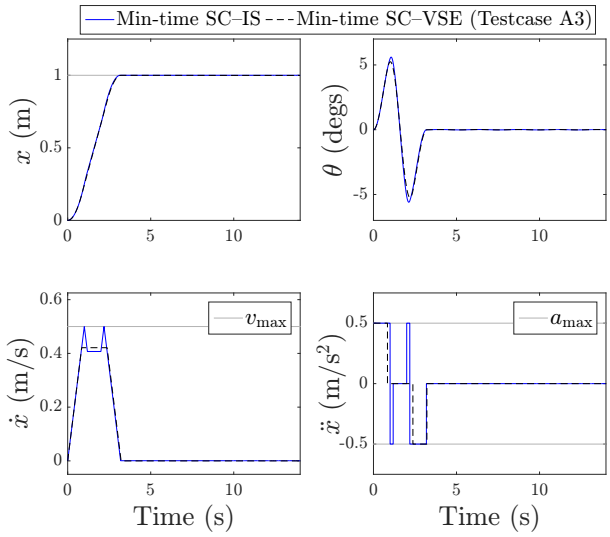


Figure 10: Simulation result of the minimum-time SC–IS and its comparison with the minimum-time SC–VSE in testcase A3.

Simulation results of the minimum-time SC–IS are depicted in Figs. 8–10 for all testcases. Like the SC–VSE case, the total transfer time for the SC–IS scheme decreases when the values of $v_{\max}$ and $a_{\max}$ increase. The vibration is suppressed at the end of the transportation. All of the actuator limits are strictly adhered to. Hence, the effectiveness of the minimum-time SC–IS is verified.

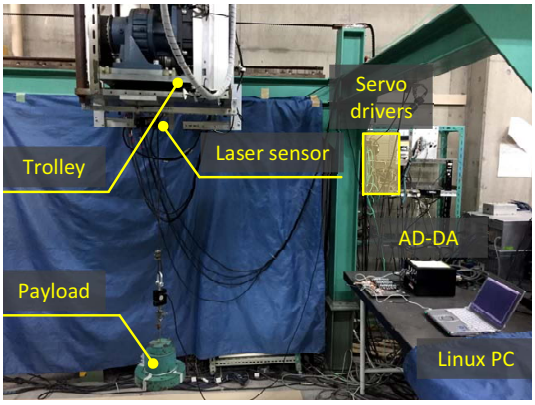


Figure 11: Structure of the experimental overhead crane system.

4.3. A brief comparison between SC–IS and SC–VSE

At the first glance, the SC–IS scheme seems to be a generalization of the SC–VSE. However, it is not true. As a counterexample, no shape family (in Table 4) of the SC–IS can produce the SC–VSE in the following configuration: $t_1 + t_2 = 2T$ (Case II), $v_{\max} = 0.5$ m/s, $a_{\max} = 0.05$ m/s^2, $l = 1.4$ m, and $x_r = 1$ m. Thus, the SC–IS does not encompass the SC–VSE in general, namely that they are complementary in relation.

Testcases A1–A2 and testcase A3, respectively, are the instances when the minimum-time SC–VSE results in a faster and longer transferring time compared to the minimum-time SC–IS (see Table 5 and Figs. 8–10). Therefore, in practice, among SC–VSE and SC–IS, it is rational to pick the one which provides a faster transferring time. This is the ultimate choice presented in this paper. Considering the case that infinite actuator limits are permitted ($v_{\max} \to +\infty$ and $a_{\max} \to +\infty$), it is easy to see that the position baseline S-curve command of the SC–IS scheme comes close to a step function to realize the minimum-time solution. In such a circumstance, the minimum transferring time of the SC–IS scheme approaches $T/2$, where the delay $T/2$ is caused by the ZV input shaper. Therefore, the SC–IS is twice as fast as the SC–VSE when infinite actuator limits are allowed. Thus, the SC–IS will be ultimately chosen.

Remark 2. Since both SC–VSE and SC–IS schemes only utilize the rope length l in their design process, the robustness to parametric uncertainties is not an issue. This can be explained by the fact that the rope length can be easily measured in practice.

Remark 3. The linearized model of a three-dimensional overhead crane system can be decoupled into two independent linear subsystems in x and y directions (Cho and Lee, 2000). Each of them is identical to the model (3), which is used in the designs of the minimum-time ZV S-curve commands. Therefore, the proposed schemes (SC–VSE and SC–IS) can be straightforwardly applied on a three-dimensional overhead crane system.

5. Experimental results

The proposed motion planning schemes are now implemented on a real overhead crane system, which is illustrated in Fig. 11. The actuator is a YASKAWA USAFED-09C2L1K AC servo motor operating in position control mode. MATLAB/Simulink is utilized to compose the program and build the executing file via Real Time Workshop. A Linux PC operates the executing file in real time with a sampling period of 10 ms through Real Time Application Interface (RTAI). Moreover, a pair of laser sensors KEYENCE VG-035 is used to measure the swing angle. Note that, the planned acceleration profile of the minimum-time SC–VSE (or SC–IS) is integrated twice to

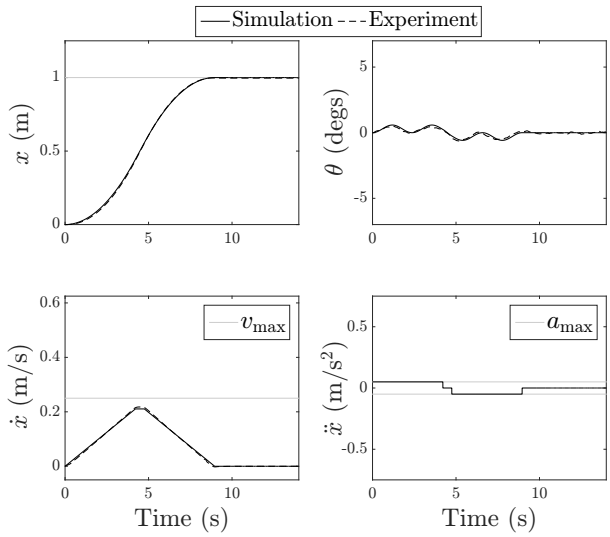


Figure 12: Experimental result in testcase A1 (the minimum-time SC–VSE is chosen). Simulation and experimental results are in a complete agreement.

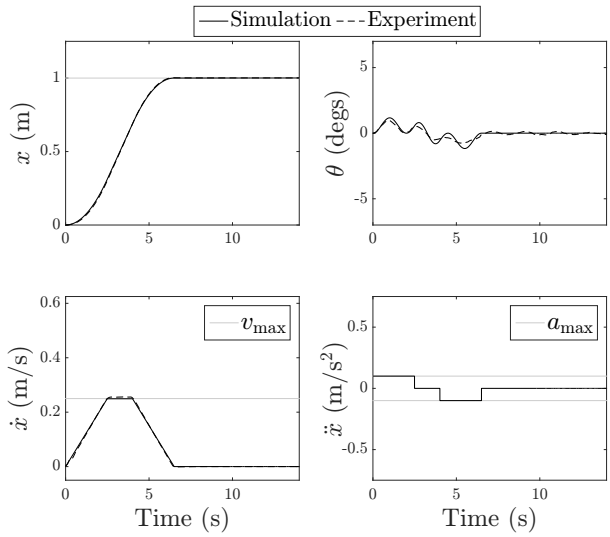


Figure 13: Experimental result in testcase A2 (the minimum-time SC–VSE is chosen). Simulation and experimental results are in a complete agreement.

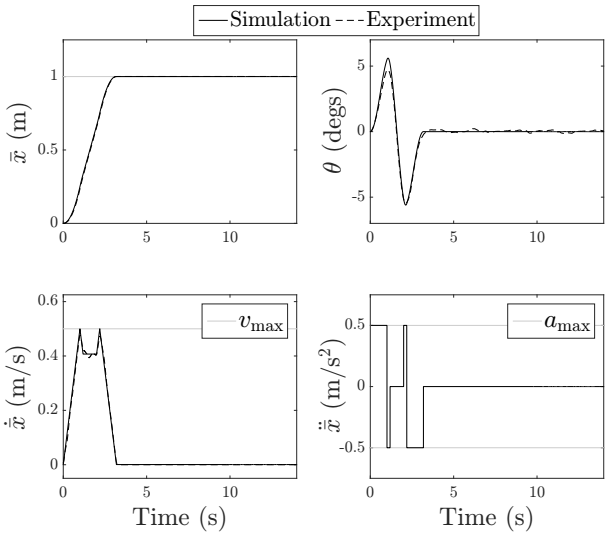


Figure 14: Experimental result in testcase A3 (the minimum-time SC–IS is chosen). Simulation and experimental results are in a complete agreement.

obtain a reference in the position level. The result is the actual feedforward command feeding to the servo driver to drive the trolley motion via a built-in motor's position controller provided by the motor manufacturer.

Three testcases A1–A3 (see Table 1) are employed to compute the minimum-time SC–VSE and the minimum-time SC–IS. Following the discussion in Section 4.3, among SC–VSE and SC–IS, the one which produces a smaller transferring time will be chosen. Hence, based on the computational results in Table 5, the minimum-time SC–VSE is chosen for testcases A1 and A2, and the minimum-time SC–IS is selected for testcase A3.

Figs. 12–14 depict the experimental results. In all testcases, it can be seen that the trolley precisely reaches the desired destination while the vibration is nearly suppressed. Moreover, the maximum velocity and maximum acceleration constraints of the actuator are strictly complied with. The experimental results strongly agree with the simulation results, which verifies the correctness of the entire modeling and control design process. Therefore, the effectiveness of the proposed method is strongly confirmed. Note that, there are slight differences between the simulations and the experimental results due to the following: i) some physical factors (e.g. air friction, payload's inertia) are neglected in the model of the overhead crane system, ii) the actual rope lengths in practice are slightly different from those used in the simulations because of small measurement errors, and iii) small tracking errors of the computed position commands exist in the experiments (see subfigure x of Figs. 12–14). Nevertheless, the differences are minor and the performances of the proposed minimum-time ZV S-curve commands have been demonstrated. Thus, such an issue is not critical.

6. Conclusions and future work

In this paper, minimum-time S-curve commands were proposed to obtain vibration-free transportation of an overhead crane system. In order to build the vibration suppression capability, two techniques were systematically established, on the basis of a parameterized baseline S-curve command. The first method directly embedded the terminal conditions of vibration-free transportation into the baseline function, whereas the second method employed the input shaping methodology. Minimum-time solutions were formulated for both schemes by solving constrained (discrete) nonlinear optimization problems, in which maximum velocity and maximum acceleration of the actuator were elaborately considered. A comparison with related studies verified the advantage of the proposed minimum-time S-curve command in terms of transfer time, and thus the transportation productivity was enhanced. It was also shown that the online command generation is feasible with the proposed schemes. Both simulation and experimental results were provided to clarify the effectiveness of the proposed methods and they are in a complete agreement. Future studies are two-fold. First, the results of this paper will be extended to a three-dimensional crane with a time-varying rope length and state constraints. Second, the nonlinear transverse flexibility of the rope, which appears when the payload is very light and/or when disturbances (e.g. wind) exert on the cable, will be considered in the control design process.

Appendix A. Proof of Theorem 1

This appendix provides the proof of Theorem 1. Based on the condition of the pivotal index $\bar{k}_1$, the proof is divided into two cases: $\max\Delta(1) \neq x_r/(Ta_{\max})$ and $\max\Delta(1) = x_r/(Ta_{\max})$ as follows.

A. *Case 1:* $\max\Delta(1) \neq x_r/(Ta_{\max})$

Since $\max\Delta(1) \neq x_r/(Ta_{\max})$, it is true that either $\max\Delta(1) = T$ or $\max\Delta(1) = x_r/v_{\max}$. In the case $\max\Delta(1) = T$, it holds that $T \geq x_r/v_{\max}$ and $T \geq x_r/(Ta_{\max})$. Therefore, $x_r/v_{\max} \leq T < 2T < \ldots$ and $\ldots < x_r/(2Ta_{\max}) < x_r/(Ta_{\max}) \leq T < 2T < \ldots$. Thus, $\max\Delta(k_1) = k_1T$, which by Lemma 1 results in

$$t_f^\dagger(k_1) = k_1T + \max\Delta(k_1) = k_1T + k_1T = 2k_1T. \quad \text{(A.1)}$$

From (A.1), since $k_1 \in \mathbb{N}^+$, $t_f^\dagger(k_1 = 1) < t_f^\dagger(k_1 = 2) < t_f^\dagger(k_1 = 3) < \ldots$. Consequently, $k_1^\star = 1$.

For the remaining case, namely $\max\Delta(1) = x_r/v_{\max}$, the result $x_r/v_{\max} \geq x_r/(Ta_{\max})$ holds. As a consequence, $x_r/v_{\max} \geq x_r/(Ta_{\max}) > x_r/(2Ta_{\max}) > \ldots$. Therefore, $\max\Delta(k_1) = \max\{k_1T,\, x_r/v_{\max}\}$ can be obtained. From that, for every $n \in \mathbb{N}^+$

$$\begin{aligned} &t_f^\dagger(k_1 = 1+n) - t_f^\dagger(k_1 = 1) \\ &= nT + \max\left\{(1+n)T, \frac{x_r}{v_{\max}}\right\} - \frac{x_r}{v_{\max}}. \end{aligned} \quad \text{(A.2)}$$

From (A.2), if $\max\{(1+n)T,\, x_r/v_{\max}\} = x_r/v_{\max}$, then $t_f^\dagger(k_1 = 1+n) - t_f^\dagger(k_1 = 1) = nT > 0$. Otherwise, $\max\{(1+n)T,\, x_r/v_{\max}\} = (1+n)T$, and the result $t_f^\dagger(k_1 = 1+n) - t_f^\dagger(k_1 = 1) = nT + (1+n)T - x_r/v_{\max}$ holds. But in this circumstance, $\max\{(1+n)T,\, x_r/v_{\max}\} = (1+n)T$, thus $(1+n)T \geq x_r/v_{\max}$; therefore, $t_f^\dagger(k_1 = 1+n) - t_f^\dagger(k_1 = 1) \geq nT > 0$. Consequently, it is concluded that when $\max\Delta(1) = x_r/v_{\max}$, the result $k_1^\star = 1$ is also true. Summarizing from the above arguments, $k_1^\star = 1$ whenever $\max\Delta(1) \neq x_r/(Ta_{\max})$.

B. Case 2: $\max\Delta(1) = x_r/(Ta_{\max})$

Flow of the proof for this case is briefly explained as follows. First, it will be proven that the local minimum transferring times $t_f^\dagger(k_1)$ (see Lemma 1) is a strictly decreasing sequence with respect to k_1 when $k_1 \leq \bar{k}_1 - 1$. Second, it will be shown that the sequence $t_f^\dagger(k_1)$ strictly increases for $\bar{k}_1 \leq k_1 < +\infty$. Finally, among $t_f^\dagger(k_1 = \bar{k}_1 - 1)$ and $t_f^\dagger(k_1 = \bar{k}_1)$, the smaller one is the global minimum.

1). Subcase 1: $k_1 \leq \bar{k}_1 - 1$ $(k_1 \in \mathbb{N}^+)$

By definition (see Theorem 1)

$$\bar{k}_1 - 1 = \max\left\{k_1 \in \mathbb{N}^+ : \max\Delta(k_1) = \frac{x_r}{k_1 T a_{\max}}\right\}. \tag{A.3}$$

From (A.3), it can be seen that $\bar{k}_1 \geq 2$. Moreover, $\max\Delta(\bar{k}_1 - 1) = x_r/\left((\bar{k}_1 - 1)Ta_{\max}\right)$. From this, it can be obtained that

$$\frac{x_r}{\left(\bar{k}_1 - 1\right)Ta_{\max}} \geq \frac{x_r}{v_{\max}} \tag{A.4}$$

$$\frac{x_r}{\left(\bar{k}_1 - 1\right)Ta_{\max}} \geq \left(\bar{k}_1 - 1\right)T. \tag{A.5}$$

The inequality (A.4) results in $x_r/(Ta_{\max}) > x_r/(2Ta_{\max}) > \ldots > x_r/\left((\bar{k}_1 - 1)Ta_{\max}\right) \geq x_r/v_{\max}$. Furthermore, (A.5) gives $x_r/(Ta_{\max}) > x_r/(2Ta_{\max}) > \ldots > x_r/\left((\bar{k}_1 - 1)Ta_{\max}\right) \geq (\bar{k}_1 - 1)T > (\bar{k}_1 - 2)T > \ldots > 2T > T$. Therefore, $\max\Delta(k_1) = x_r/(k_1 Ta_{\max})$. Since $k_1 = 0$ is not a legitimate index, $t_f^\dagger(k_1 = 0) = +\infty$ is defined for the sake of convenience. Therefore, $t_f^\dagger(k_1) - t_f^\dagger(k_1 - 1)\Big|_{k_1=1} < 0$. In the case that $k_1 \geq 2$, by using Lemma 1, it yields

$$\begin{aligned} &t_f^\dagger(k_1) - t_f^\dagger(k_1 - 1) = \\ &k_1 T + \frac{x_r}{k_1 T a_{\max}} - (k_1 - 1)T - \frac{x_r}{(k_1 - 1)Ta_{\max}}. \end{aligned} \tag{A.6}$$

The result of (A.6) can be simplified to

$$t_f^\dagger(k_1) - t_f^\dagger(k_1 - 1) = T - \frac{x_r}{k_1(k_1 - 1)Ta_{\max}}. \tag{A.7}$$

Since $\max\Delta(k_1) = x_r/(k_1 Ta_{\max})$, $\forall k_1 \leq \bar{k}_1 - 1$, it holds that $x_r/(k_1 Ta_{\max}) \geq k_1 T$; therefore, $x_r/(k_1 Ta_{\max}) > (k_1 - 1)T$, so $T < x_r/\left(k_1(k_1 - 1)Ta_{\max}\right)$. Hence, (A.7) results in $t_f^\dagger(k_1) - t_f^\dagger(k_1 - 1) < 0$. In summary

$$t_f^\dagger(k_1) < t_f^\dagger(k_1 - 1),\ \forall k_1 \leq \bar{k}_1 - 1\ \left(k_1 \in \mathbb{N}^+\right). \tag{A.8}$$

The result in (A.8) implies that the sequence $t_f^\dagger(k_1)$ of local minimum transferring times strictly decreases when $k_1 \leq \bar{k}_1 - 1$.

2). Subcase 2: $\bar{k}_1 \leq k_1 < +\infty$ $(k_1 \in \mathbb{N}^+)$

From the definition of $\bar{k}_1$ (see (A.3)), it is true that $\max\Delta(\bar{k}_1) \neq x_r/(\bar{k}_1 Ta_{\max})$, namely $\max\Delta(\bar{k}_1) = \max\left\{\bar{k}_1 T,\, x_r/v_{\max}\right\}$. From this

$$\frac{x_r}{\bar{k}_1 Ta_{\max}} \leq \bar{k}_1 T \text{ and } \frac{x_r}{\bar{k}_1 Ta_{\max}} \leq \frac{x_r}{v_{\max}}. \tag{A.9}$$

From (A.9), the following results are obtained:

$$\ldots < \frac{x_r}{(\bar{k}_1 + 1)Ta_{\max}} < \frac{x_r}{\bar{k}_1 Ta_{\max}} \leq \bar{k}_1 T < (\bar{k}_1 + 1)T < \ldots \tag{A.10}$$

$$\ldots < \frac{x_r}{(\bar{k}_1 + 1)Ta_{\max}} < \frac{x_r}{\bar{k}_1 Ta_{\max}} \leq \frac{x_r}{v_{\max}}. \tag{A.11}$$

It can be immediately seen from (A.10) and (A.11) that

$$\max\Delta(k_1) = \max\left\{k_1 T,\, \frac{x_r}{v_{\max}}\right\},\ \forall k_1 \geq \bar{k}_1. \tag{A.12}$$

Therefore, $\forall k_1 \geq \bar{k}_1$ $(k_1 \in \mathbb{N}^+)$, by using Lemma 1

$$\begin{aligned} &t_f^\dagger(k_1 + 1) - t_f^\dagger(k_1) = T \\ &+ \max\left\{(k_1 + 1)T,\, \frac{x_r}{v_{\max}}\right\} - \max\left\{k_1 T,\, \frac{x_r}{v_{\max}}\right\}. \end{aligned} \tag{A.13}$$

From (A.13), two possibilities can be classified: $\max\{k_1 T,\, x_r/v_{\max}\} = k_1 T$ and $\max\{k_1 T,\, x_r/v_{\max}\} = x_r/v_{\max}$. For the first possibility, that is to say, $\max\{k_1 T,\, x_r/v_{\max}\} = k_1 T$, the inequality $x_r/v_{\max} \leq k_1 T < (k_1 + 1)T$ holds, thus $\max\{(k_1 + 1)T,\, x_r/v_{\max}\} = (k_1 + 1)T$. Therefore, (A.13) becomes

$$t_f^\dagger(k_1 + 1) - t_f^\dagger(k_1) = T + (k_1 + 1)T - k_1 T = 2T > 0. \tag{A.14}$$

For the second possibility, that $\max\{k_1 T,\, x_r/v_{\max}\} = x_r/v_{\max}$, (A.13) can be simplified to

$$\begin{aligned} &t_f^\dagger(k_1 + 1) - t_f^\dagger(k_1) = \\ &T + \max\left\{(k_1 + 1)T,\, \frac{x_r}{v_{\max}}\right\} - \frac{x_r}{v_{\max}}. \end{aligned} \tag{A.15}$$

From (A.15), it is not difficult to show that

$$t_f^\dagger(k_1 + 1) - t_f^\dagger(k_1) \geq T > 0. \tag{A.16}$$

From (A.14) and (A.16), the conclusion is

$$t_f^\dagger(k_1+1) > t_f^\dagger(k_1),\ \forall k_1 \geq \bar{k}_1\ \left(k_1 \in \mathbb{N}^+\right). \tag{A.17}$$

The result of (A.17) implies that the local minimum transferring times $t_f^\dagger(k_1)$ is a strictly increasing sequence with respect to k_1 when $\bar{k}_1 \leq k_1 < +\infty$.

Summarizing the results in (A.8) and (A.17) gives

$$\begin{cases} t_f^\dagger(k_1) < t_f^\dagger(k_1-1), & \forall k_1 \leq \bar{k}_1 - 1 \\ t_f^\dagger(k_1) < t_f^\dagger(k_1+1), & \forall k_1 \geq \bar{k}_1 \end{cases},\ k_1 \in \mathbb{N}^+. \tag{A.18}$$

Therefore, the global optimal index of the mixed nonlinear–integer program (19) is the one among $\bar{k}_1 - 1$ and $\bar{k}_1$, which produces a smaller transferring time. Using the result of Lemma 1 results in

$$\begin{aligned} &t_f^\dagger(k_1 = \bar{k}_1) - t_f^\dagger(k_1 = \bar{k}_1 - 1) = \\ &T + \max\Delta(\bar{k}_1) - \max\Delta(\bar{k}_1 - 1). \end{aligned} \tag{A.19}$$

By utilizing the definition of μ in Theorem 1 and (A.19), the conclusion is that when $\mu \geq 0$, namely $t_f^\dagger(k_1 = \bar{k}_1) \geq t_f^\dagger(k_1 = \bar{k}_1 - 1)$, the global optimal index of (19) is $k_1^\star = \bar{k}_1 - 1$. Otherwise, namely $\mu < 0$, the result $k_1^\star = \bar{k}_1$ holds. This completes the proof.

Appendix B. Proof of Theorem 2

The first step of the proof is to find the local optimal solutions of (23) for each $k_2 \in \mathbb{N}^+$ satisfying $k_2 \geq \max\left\{x_r/(Tv_{\max}),\ (1/T)\sqrt{x_r/a_{\max}}\right\}$. In the second step, the local minima provided by the first step (which are infinite in number) will be compared to each other to justify the global optimum.

The first step can be done as follows. For a fixed $k_2 \in \mathbb{N}^+$ satisfying $k_2 \geq \max\left\{x_r/(Tv_{\max}),\ (1/T)\sqrt{x_r/a_{\max}}\right\}$, by introducing a new variable $\tilde{z} = x_r/(\tilde{a}k_2T)$, (23) can be transformed into the following simple linear optimization problem:

$$\begin{aligned} \min_{\tilde{z}\in\mathbb{R}^+} \quad & \tilde{t}_f = k_2T + \tilde{z} \\ \text{s.t.} \quad & \frac{x_r}{k_2Ta_{\max}} \leq \tilde{z} \leq k_2T. \end{aligned} \tag{B.1}$$

It is easy to show that the minimizer $\tilde{z}^\dagger$ and the minimum cost function $\tilde{t}_f^\dagger$ of (B.1) are $\tilde{z}^\dagger(k_2) = x_r/(k_2Ta_{\max})$ and $\tilde{t}_f^\dagger(k_2) = k_2T + \tilde{z}^\dagger(k_2)$. Using the relation $\tilde{z} = x_r/(\tilde{a}k_2T)$ results in $\tilde{a}^\dagger(k_2) = a_{\max}$ and $\tilde{t}_f^\dagger = k_2T + x_r/(k_2Ta_{\max})$.

Now, the second step can be proceeded by solving the following optimization problem:

$$\begin{aligned} \min_{k_2\in\mathbb{N}^+} \quad & \tilde{t}_f^\dagger(k_2) = k_2T + \frac{x_r}{k_2Ta_{\max}} \\ \text{s.t.} \quad & k_2 \geq \max\left\{\frac{x_r}{Tv_{\max}},\ \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}}\right\}. \end{aligned} \tag{B.2}$$

Considering a function $\tilde{t}_f^\dagger(y)$ where $y \in \mathbb{R}^+$, $y \geq \max\left\{x_r/(Tv_{\max}),\ (1/T)\sqrt{x_r/a_{\max}}\right\}$. Taking the first derivative of $\tilde{t}_f^\dagger(y)$ with respect to y results in

$$\frac{d\tilde{t}_f^\dagger(y)}{dy} = T - \frac{x_r}{y^2Ta_{\max}}. \tag{B.3}$$

However, the feasible region of y gives $y \geq (1/T)\sqrt{x_r/a_{\max}}$, thus $T \geq x_r/\left(y^2Ta_{\max}\right)$. By applying this result on (B.3), $d\tilde{t}_f^\dagger(y)/dy \geq 0$ can be obtained, which means that the function $\tilde{t}_f^\dagger(y)$ monotonically increases with respect to y. By performing the substitution $y = k_2$, it can also be concluded that the function $\tilde{t}_f^\dagger(k_2)$ monotonically increases with respect to k_2. Therefore, the global optimal index $k_2^\star$ of (B.2) is the minimum integer satisfying $k_2^\star \geq \max\left\{x_r/(Tv_{\max}),\ (1/T)\sqrt{x_r/a_{\max}}\right\}$. In the set notation, the above result is equivalent to $k_2^\star = \min\{k_2 \in \mathbb{N}^+ : \max\beta(k_2) = k_2\}$. The proof is therefore completed.

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